

# Research as an Imperative for Clinical Excellence

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I am very grateful to our Congress of Neurological Surgeons (CNS) President, Alex Khalessi, and the CNS Executive Board for giving me the chance to share with you, a little bit about my own journey that shaped me into the neurosurgeon and neurosurgeon-scientist that I am today.

## CAREER FORGED BY FAMILY

To understand my background, we need to go back to 1949. My parents arrived in the United States from a camp of displaced persons in Germany after surviving the Holocaust.

This photograph (Figure 1A) captures a happy moment with my older brother Steven, who is now a Professor of Neurosurgery at the University of Pennsylvania, and my parents, Adele and Jacob Brem, two remarkable individuals. Just at the age of 20 years, they did not speak English, had no high school education, and had lost their entire families to the Nazi Holocaust. However, they carried with them an incredible resilience and joy for life. They rebuilt their family, and our lives were filled with joy.

Despite all they had endured—the unimaginable loss of parents, siblings, and everyone close to them—they came to America with hope, ambition, and a strong desire to build a better future. Figure 1B shows my parents, with their four children—my older brother Steven; myself; my sister Shari who is a periodontist; and my younger brother Harold, who is a general surgeon.

Our upbringing was grounded in the values of our parents instilled in us. These values laid the foundation for how I approach medicine today. First, there was our heritage that they imbued us with. There was a passion for learning and a respect for educators and respect for how important our teachers were, at every stage of our lives.

Another principle that shaped me is the Jewish concept of *TIKUN OLAM*, the idea that we are all responsible for making the world a better place. This belief has been central to everything I do. We were taught to strive to leave every situation better than we found it.

There was also a very deep love of mankind and fellow human beings. This is despite the horrors of what our parents had survived and the worst of humanity that they had seen with concentration camps and death camps. Instead, we were brought up with joy, compassion, and love.

There is a biblical expression, “*Ve-ahavta Le-reaskha kamokha*” Leviticus 19:18, which means *love your fellow man as yourself*.

We were imbued with a very deep sense of family, and this love of family also shaped me as a physician. Finally, they passed on a profound appreciation for life itself. There is a Talmudic expression that is enshrined on the entrance hall of the Johns Hopkins Hospital, “*if you save one life, it’s as if you’ve saved the whole world.*” This has been a driving force behind my career in medicine.

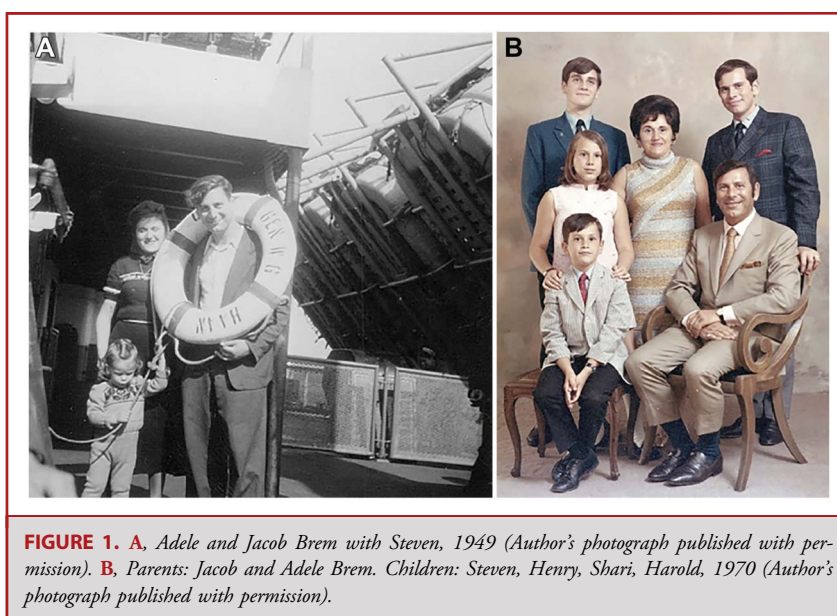
In terms of understanding who I am, the most important person in my life is my wife Rachel (Figure 2A). We started dating when she was at the age of 17 years at Brandeis. We were married in 1978, just six months before I started my internship at the Peter Bent Brigham Hospital, as Rachel was completing her premedical courses. The second photograph (Figure 2B) captures us at the same spot at the Brigham 40 years later—I do not see any significant changes! Our life together has greatly defined who I am, both personally and professionally.

Family has always been a big part of our life, and because of that wonderful 47-year journey with Rachel, we have three incredible daughters, three sons-in-law, and 10 amazing grandchildren, and we are thrilled that they all live in the Baltimore-Washington area (Figure 2C).

As a physician, the deep love I experience within my family gives me a deep empathy for my patients and their families. I understand, firsthand, the profound emotional journey that families undergo when a loved one is ill, and this connection helps me care for them with greater compassion.

## IMPACT AND IDEAL MENTORSHIP

When I chose a career in medicine, the word that resonated most with me was IMPACT. I wanted to maximize my ability to



**FIGURE 1.** A, Adele and Jacob Brem with Steven, 1949 (Author's photograph published with permission). B, Parents: Jacob and Adele Brem. Children: Steven, Henry, Shari, Harold, 1970 (Author's photograph published with permission).

improve lives. I wanted to choose a path where I could have the greatest impact, and neurosurgery was that path. The ability to change lives so dramatically through our work as neurosurgeons is extraordinary. Along the way, I have been inspired by incredible mentors—both in medicine and in science. The beauty of medicine lies in the seamless integration of science and humanity and that has been endlessly fascinating to me.

There were books that I read as a college student at New York University that were very inspirational and very important to me in building who I wanted to become. I was deeply moved by Christiaan Barnard who, in South Africa, performed the first human heart transplant taking advantage of incredible science and medicine and breaking barriers,<sup>1</sup> the Doctors Mayo, a father and two brothers, who revolutionized American surgery,<sup>2</sup> and then fatefully, because I am the Harvey Cushing Professor at Johns Hopkins, that in college I was so inspired by both Cushing's biography<sup>3</sup> and by his own profound writings.<sup>4</sup> Scientifically, I was very inspired by Pasteur and the incredible impact that his scientific discoveries had on mankind.<sup>5</sup> René Dubos<sup>6</sup> writes about the enthusiasm, which he calls "A God within," that inspires science.

In preparing this presentation, I revisited a lecture by Harvey Cushing from 1929. He highlighted the importance of blending science with clinical practice, suggesting that ideal practitioners engage in both fields. "There are among doctors, the scientists who may never see a patient, and the practitioners, whose work is based on the knowledge that comes from the many scientific laboratories, and who indeed, if spurred by curiosity and given the proper training, may profitably engage in scientific research and medical practice at the same time, which is the ideal combination"<sup>4</sup> This quote from Harvey Cushing describes how important it is to have both science and medicine in the same person. Interestingly, I could

not resist sharing another quote as well from the same lecture, Cushing states: "Every generation is prone to over-value its own accomplishments. And to feel, that it has lived through the most eventful era in history. So, it's music to our ears to be told that the recent half century [1920s], has seen the greatest progress of all!" and Cushing goes on to say, "Not everyone is entirely happy about it..." I think nothing's changed on that front!

Mentorship is so critically valuable, it is how we pass on our experience, knowledge, and our approaches, but it is not unidirectional. I think that the mentee and the mentor both invest major efforts, but the outcome is phenomenal and transformative!

My most influential mentor has been Dr Judah Folkman (Figure 3A and 3B), an incredible surgeon scientist, teacher, and humanist, to whom I am incredibly indebted.<sup>7</sup> Years later, when he was a visiting professor at Hopkins, I tried to express how grateful I was to him for what he had imparted to me, and I said I wish I could somehow do something for him. He was the kindest, gentlest person in the world, so I was surprised that he said:

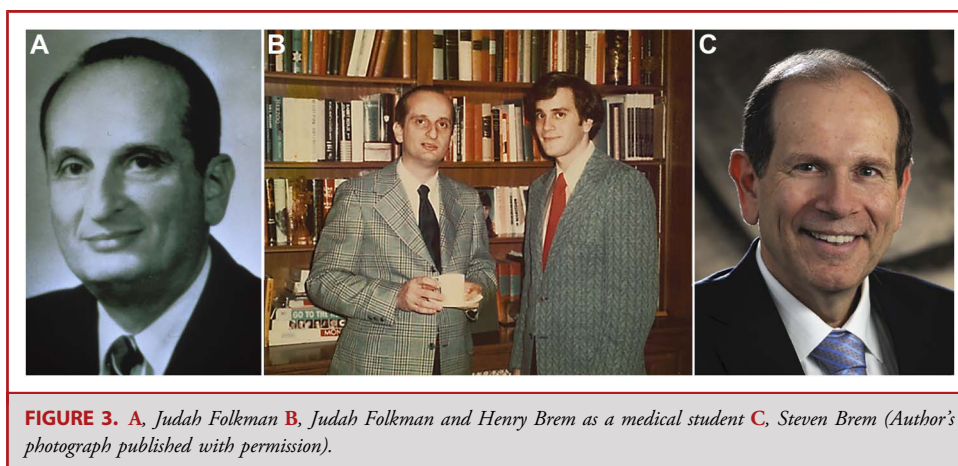
"No, there's nothing you can do for me... whatever you think it is that I've done for you, give it to your students, do the same." That's always been my ideal, I don't think that I've come close to what he did for me, but that has been my aspiration.

My brother Steven Brem (Figure 3C) has been a lifelong influence on me—inspiring me to pursue medicine as a career and to choose neurosurgery as a passion. His love of science and medicine was inspirational, and indeed, it was Steven who introduced me to Judah Folkman.

Other major mentors for me who have INSPIRED my development as a surgeon scientist and had a substantial impact on me are as follows: Arnall Patz<sup>8</sup> who revolutionized ophthalmology







**FIGURE 3.** A, Judah Folkman B, Judah Folkman and Henry Brem as a medical student C, Steven Brem (Author's photograph published with permission).

and won the Lasker Award for curing the principal cause of neonatal blindness at the time, retrolental fibroplasia.

Donlin M. Long who revolutionized the understanding of brain tumors in his PhD work and nurtured an environment at Hopkins, where a surgeon truly could devote himself to the laboratory and to being a master surgeon in the operating room,<sup>9</sup> and Michael Colvin who mentored me in oncology and pharmacology, so I could answer the questions that I was trying to address for brain tumors.<sup>10</sup>

Solomon (Sol) Snyder, a great neuroscientist, has been a major influence, friend, and mentor in so many ways and in helping build up some of the work that I am going to briefly discuss. Finally, my closest friend Robert (Bob) Langer (Figure 4), with whom I have worked as students in Judah Folkman's laboratory and shared Dr Folkman as a mentor. Bob has performed extraordinary work and changed the world through his research. Bob Langer's work is not only critical to my research in brain tumors, but he is also a founder of Moderna, who developed the COVID-19 vaccine that has saved millions of lives!<sup>11</sup>

## RESEARCH

Reflecting on my journey, I think about the various phases that shaped my focus in medicine. At the age of 18 years, I had the incredible opportunity to work in the Columbia microbiology

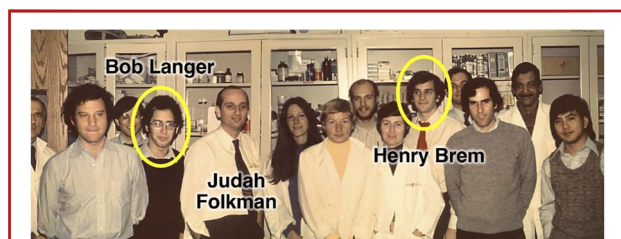
laboratory with Herbert Rosenkrantz, where I investigated the mutagenicity of alkylating agents, while also gaining hands-on experience in the operating rooms at Columbia. During that summer, I published two articles on 1,2 di-bromoethane as a carcinogenic agent and its mechanisms of action.<sup>12,13</sup> This experience taught me a valuable lesson: even with limited scientific background, if you narrow your research question down deeply enough, you can become an expert in that specific area and make meaningful contributions. This principle has guided many of my scientific endeavors.

When I had the fortune of working with Judah Folkman, I read his work and his hypothesis that angiogenesis was a critical control point in tumor growth. While I was taking embryology in college, I came across the observation that cartilage is vascularized in the fetus and then it loses its vascularity. Having read so much about the theory of antiangiogenesis and talking to Dr Folkman about it, I suggested, based on that observation in nature, that we will investigate if cartilage has an active inhibitor for angiogenesis at the point that it loses its vascularity (Figure 5).<sup>14</sup>

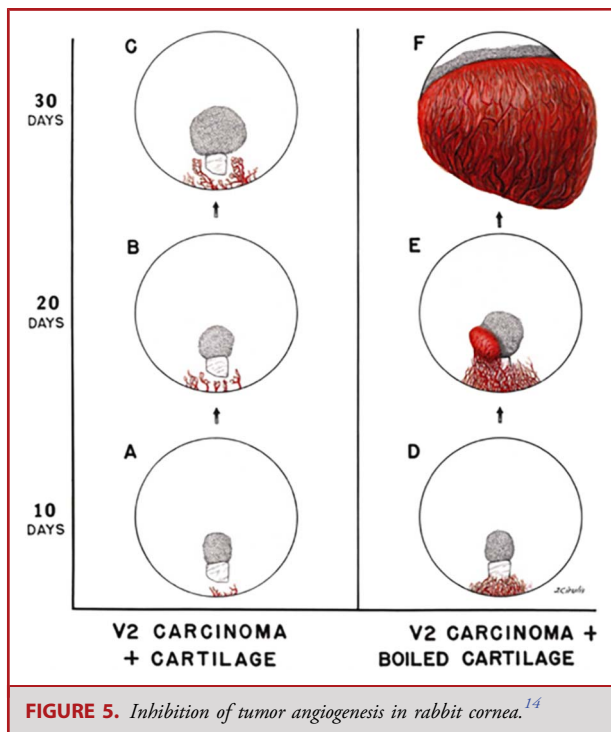
We spent a year working with Judah Folkman on this, and this was the key finding that cartilage in fact had an inhibitor for angiogenesis.<sup>14,15</sup> Antiangiogenesis has had a profound impact in treating ocular disease, dermatological conditions, and, of course, cancer. I am proud of my role in initiating this process and, even more importantly, the surgeons who were responsible for initiating these profound scientific and clinical advances.

Stemming out of this work, Bob Langer and Judah Folkman wrote the first article on polymers for the sustained release of proteins and macromolecules effectively creating the field of drug delivery,<sup>16</sup> and later together with David Tapper, who eventually became the chairman of pediatric surgery at the University of Washington, we explored how to make these polymers safe for human applications.<sup>17</sup>

I was very excited by science, but I was really thrilled to be in medical school! As I look back at medical school, in addition to the incredible thrill of working with Judah Folkman and others, it was really about learning and improving medicine and devoting



**FIGURE 4.** Dr Judah Folkman's laboratory. 1974 (Author's photograph published with permission).



tremendous amount of effort and time to mastering the scientific and clinical sciences, enabling me to provide the best care for my patients. Judah Folkman frequently spoke about the distinction between personal ignorance and general ignorance.

Personal ignorance, failing to know something that is known information, but you do not know it—is unacceptable because you can potentially harm a patient when it was preventable. That is intolerable. By contrast, general ignorance encompasses the scientific questions we still need to answer through research, such as how to cure glioblastomas. As a neurosurgeon, I have always felt a profound trust from my patients because they allow me to care for them and perform surgeries, placing their faith in me, essentially a stranger, to help them through some of their most vulnerable moments, and yet they trust me enough to allow me to do my best, given the best available knowledge, to help them.

We were taught that you treat people the way that you would want to be treated. However, what I have learned over time is that this rule is not an adequately high enough standard. The goal really was to understand how the patient wants to be treated and to make sure that you are meeting that expectation.

My neurosurgery residency at Columbia was really six years of just incredible devotion to learning neurosurgical anatomy, neurosurgical pathophysiology, and of course, the incredible demanding techniques that are required. These challenges consumed my entire focus during that time. When I started on the faculty at Hopkins in 1984, I knew that I had this incredible scientific background, I was fascinated by science, and that scientific discovery was very rewarding. Hence, I understood the

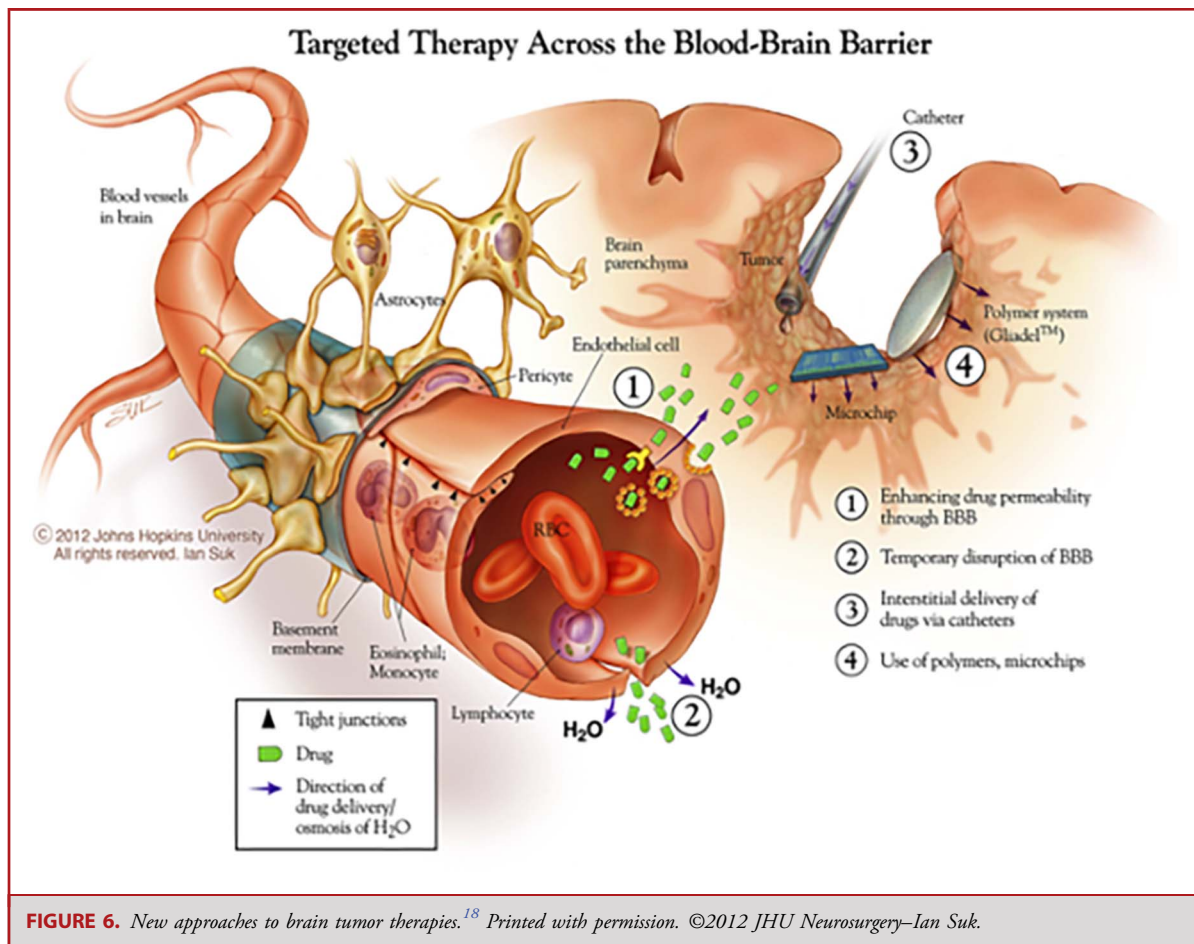
possible impact it could have. However, I also felt that I had spent seven years learning neurosurgical techniques and caring for neurosurgical patients, and I knew that clinical neurosurgery itself was exceptionally rewarding and demanding.

I felt nervous about taking care of patients independently. Could I take care of patients and do brain surgery and spine surgery? Would other physicians trust me to take care of their patients? Could I be a good doctor, not as a resident not under someone else's supervision, but on my own? So, when I started as an attending, I was very focused on building a practice and building special relationships with patients. I would wake up early before an operation, and I would restudy the anatomy and technique. I was very relieved and delighted that very quickly I gained a trustworthy reputation; that I built up a large practice; that physicians and other patients would come to me, particularly for brain tumor surgery; that people trusted me; and that I fulfilled that trust and did the best that was available in medicine at that time, but, after a little while, it was a relief that yes, I could do these things that I was taught and could achieve the best results of our specialty. Indeed, this was extraordinarily satisfying. However, I began to realize that it really was not enough. I realized that there was a huge emptiness from it because I knew that patients were coming from great distances, for example, with malignant brain tumors, and I became very close with those patients, and they trusted me. I worked with them and helped their families, and we formed these incredible bonds ..., and then they died. I realized that even with all the work that I was doing, and all the trust that they put in me, the outcome was still unacceptable. I realized that we just did not know how to do better.

When I was close with the patients, and I asked them if they felt like I let them down, the answer that I got was, "No, we understood that there was no cure for this, but we know that you are working on making it better, you are at Hopkins, you are a scientist and a neurosurgeon and that is OK. We want to be part of that process of making the outcomes better." I realized that I had a contract with those patients and a responsibility to those patients not only to do my best with the tools available but also to find ways to do better than what was expected of me. Here, I recognized that I had the opportunity and training to do so!

After an initial period of practice, there was an understanding that it just was not enough. I felt like I could operate expertly, I could do what I was trained to do and what the profession allowed, but it really was not enough, and we just needed to do more. If the goal is cure, then we basically fail! That is true in so much of medicine, extremely so when it comes to the patients with glioblastoma that we take care of, but really in so much of medicine. However, if the objective is to improve the outcomes, then we do succeed. We succeed both by our clinical expertise and compassionate, meticulous care of the patients and their families and their friends.

In speaking with my patients, I realized that even the ones that do die in a year and do have this terrible outcome... they are grateful that me and the whole team were there with them in the process and that we were always there for them, we helped them, and they were not abandoned. They understood that we do not



**FIGURE 6.** New approaches to brain tumor therapies.<sup>18</sup> Printed with permission. ©2012 JHU Neurosurgery—Ian Suk.

create the diseases, but we can truly help people deal with them, as they go through the process, but the real moral obligation, the real change that I felt had to happen, was to change those outcomes. I felt that had to happen! We had to create even incremental advances in measurable outcomes.

Another insight that I came to realize was that, to some extent, as much as I care about people, and I do deeply care, we also create psychological shields to create some distance from the pain and the suffering because it is overwhelming how much suffering some of our patients go through. We sort of think “they are the patients, at arm’s length, and that we are over here, we are the physicians and care team.” I think that is what gets us through and that is how we can go home and be with our families or go to a laboratory and do research. We can walk away from the pain and suffering that we also are involved with when we are with our patients. We get good at compartmentalizing.

However, despite that sort of psychological shield that we put up, some patients penetrate that protective shield, and they make us feel their suffering very deeply, and then, we struggle to stay clear-minded and objective. If that shield is broken, it is difficult to do some of the hard things we have to do.

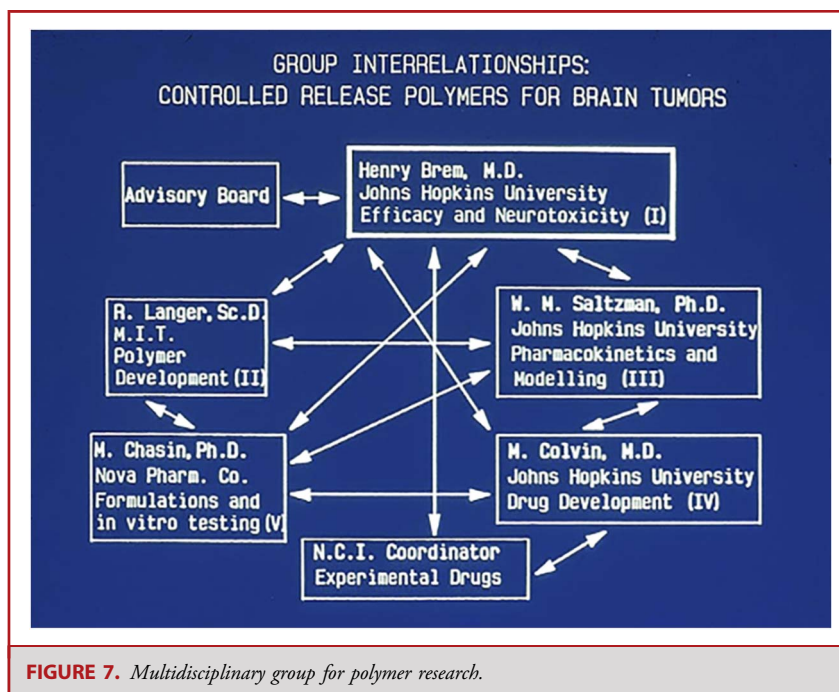
Nevertheless, feeling that suffering is a very strong motivator, and it does constantly remind us of how serious the problems those patients are facing when they come to us, to get our help. There is a moral imperative to give back for the trust those patients and their families put in us. That trust obligates us to do research and improve the inadequacies of medicine and surgery.

When it comes to choosing which way one should go in their career, I have also come to realize that, interestingly, in many ways, research as a career is a much higher risk than brain surgery. The reality is brain surgery is a body of knowledge that you learn, you study, you work at, and you master its techniques, and if you follow the rules and do what could be done, there is very little personal risk, career-wise, to the person.

On the other hand, in a research career, you could spend your whole life working on a scientific hypothesis, and work day and night, and it could turn out to be nothing! It could turn out to be an artifact; it could turn out to be meaningless. So, in fact, the risk, career-wise, is in a sense much greater in research!

If your goal is to be impactful, you could really spend a lifetime in a laboratory and not be impactful. By contrast, the consequences of





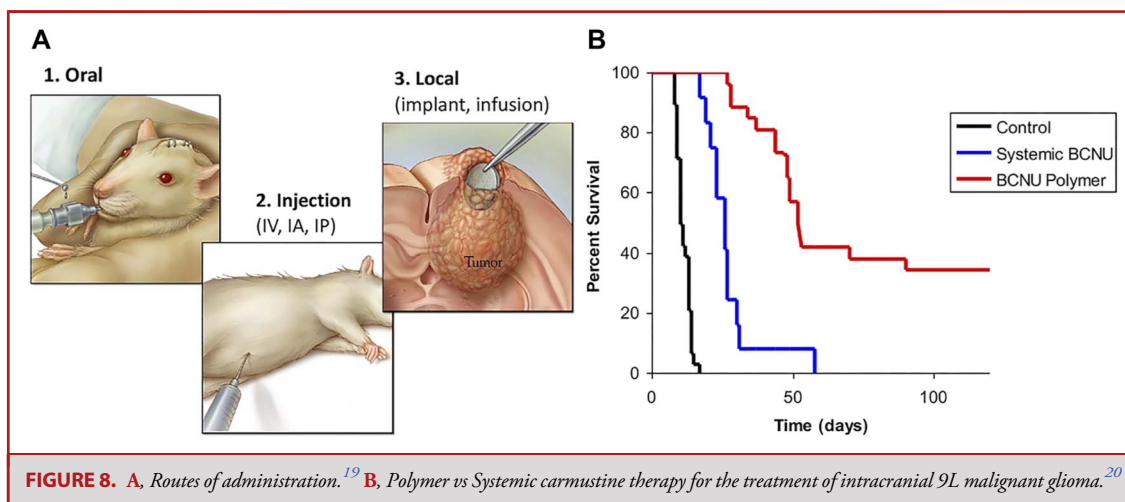
neurosurgery and what I do clinically are immediate and very great to my patients. However, I think the impact is much greater in research, and I would even say that as I look back at my own career.

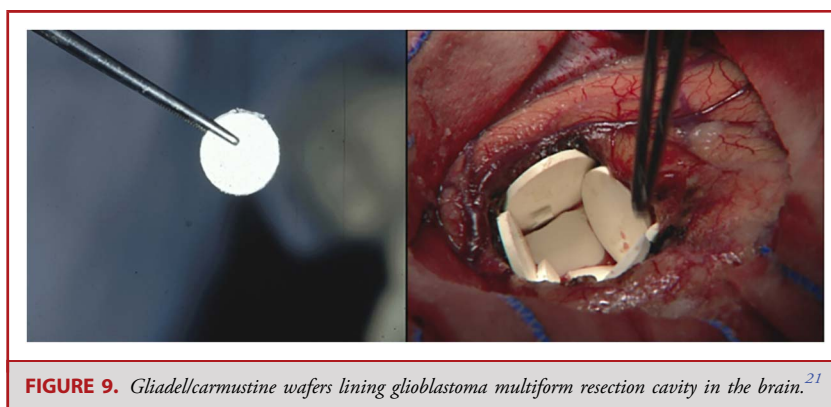
Although I have helped thousands of patients by operating on them and taking care of them..., and the positive satisfaction of surgery is enormous. I have affected many more patients in an indirect way through the research I have done!! Mentoring our trainees to follow this combined pathway has even further increased the impact through their subsequent work, which reinforces Harvey Cushing's ideal of the combination of the practitioner and researcher. Many more patients have benefited

from our discoveries and our findings than I have ever seen as an individual person. So, it is interesting how the two aspects of our careers have very different risks and benefits to them for the people who practice them.

## THERAPEUTIC DELIVERY

One of the largest efforts in brain tumors is the work we have done in delivery of therapeutics and overcoming the problem of clinical ineffectiveness of every new cancer therapy at the time.





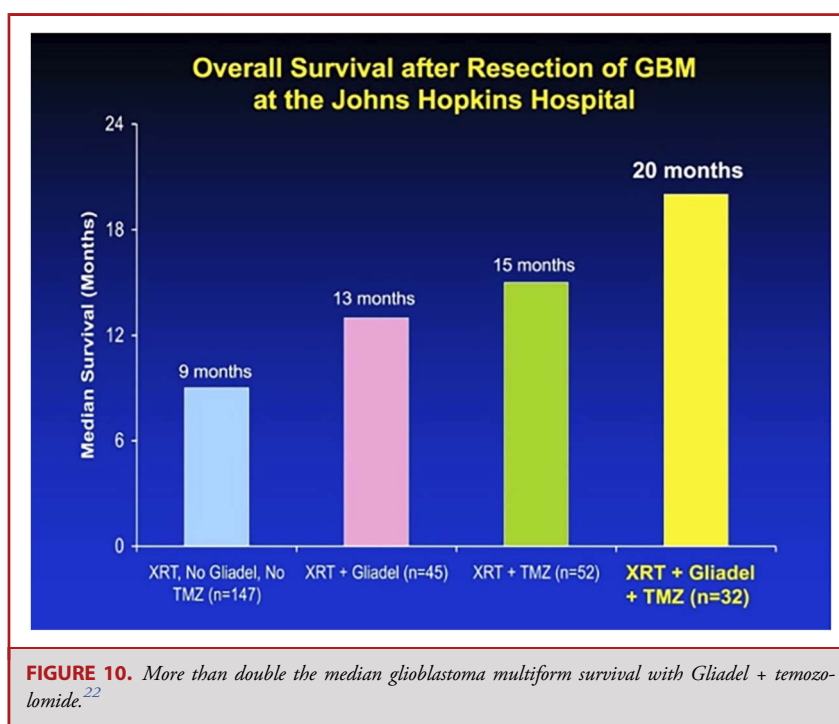
Crossing the blood-brain barrier was the big hurdle (Figure 6). We had hypothesized that maybe these therapies just were not getting to where they needed to go, so we used the polymers that had been developed at the Massachusetts Institute of Technology for drug delivery, which had never been used previously in humans.

We established a multidisciplinary group at Hopkins with an NIH funding of \$1 000 000/year at that time, to work with a pharmaceutical company, Nova Pharmaceutical Corp (founded based on Sol Snyder's neuroscience discoveries), to refine the pharmacology with biomedical engineering (Figure 7).

Dr Rafael Tamargo, currently the Walter Dandy Professor of Neurosurgery at Johns Hopkins and Director of our vascular service, worked with us as a fellow and helped develop the rat

brain tumor model (Figure 8A).<sup>20</sup> We implanted brain tumors into these rats and subsequently gave those systemic infusions or oral medications vs direct local delivery with polymers or with catheters for convection enhanced delivery. We showed that, in fact, we could, using polymer-based delivery, get a much better survival for the same drug as compared with it being delivered systemically (Figure 8B).

This is a photograph of a surgery where we have taken out the brain tumor and we line the resection cavity with carmustine wafers (Figure 9). The release of the loaded chemotherapeutic agents occurs over several weeks, and we showed that this approach was effective with a statistically significant improvement in survival (Figure 10).





## This Approach Will NOT Work Because:

- Polymers cannot be synthesized (1981)
- Polymers will react with encapsulated drugs (1983)
- These polymers are fragile (1985)
- The polymer drug system would be toxic (1987)
- Drugs would not diffuse far enough (1989)
  
- Models do not reflect clinical reality (1991)
- BCNU is a very poor drug (1993)
- FDA approval would be impossible for a polymer system (1995)
- How will it be paid for? (1997)
- Which patients will maximally benefit? (1999)
- Would the FDA broaden the indications? (2003)
- Precludes phase I studies (2005) vs. combinational benefits!
  
- Need better targeted drugs (2007)
- Need more sophisticated delivery approaches: Microchips, Ultrasound and Nano-technology (2024)

**FIGURE 11.** List of objections to innovation toward polymer therapeutic delivery.

As Bob Langer and I and many colleagues at Hopkins were developing this work and presenting it at meetings, what we found is that it was very well received. Colleagues respected the work, agreed that it was good science, but they would add, “it’ll never work though, you know that.” What was interesting is that over the years different reasons were stated as to why it would not work. Figure 11 summarizes some of the objections to innovation that were systematically overcome with data.

First, they said the polymers could not be synthesized, they would react with the drugs. Then, they said that the polymers

were fragile, they would be toxic, and the drug could not travel far enough into the brain. So, we answered and published the experiments to answer these issues.

We did not get angry, and we did not drop the idea. We just said OK; we will explore these issues, and test them, and develop them. We went back to the laboratory, and we did all these experiments, and we published, and we just systematically answered these questions. Colleagues then said models do not reflect clinical reality; anybody can cure cancer in mice! That is not what is going to happen with people.

## Harvey Cushing and Walter Dandy

“...Neurological advance had depended not only upon great workers in this specialism, like Jackson and Horsely and Sherrington, but upon a multitude of labourers in other arts and sciences the connexion of which was not at first obvious—upon Michael Faraday and Henry Bessemer, upon Edison and Crookes and Röntgen, upon Lister and Pasteur and Simpson, upon workers with the microscope and in chemistry, and so on in an endless chain. The progress from the days when a piece of flint was drilled into the skull to let out evil spirits to the present time, when a brain tumour could be detected and removed, was merely the result of the adaptation of **innumerable fragments of applied knowledge.**”

**Harvey Cushing:** *The Cavendish Lecture. Br Med J. June 24, 1922*

“Scientific accuracy must supplant guesswork in diagnosis and directing the treatment”

**W.E. Dandy:** *JAMA 77:1858-1859 (1921)*

**FIGURE 12.** Quotes from Harvey Cushing and Walter Dandy.<sup>25,26</sup>



So, we did the clinical studies. We used the same rigor that we used in the laboratory. It was very challenging to do so. However, by doing that, we showed the Food and Drug Administration (FDA) that we really had something that was beneficial.<sup>23</sup> We did a randomized prospective placebo-controlled study in 27 medical centers showing a statistically significant improvement in survival for recurrent malignant gliomas.<sup>20</sup> Nonetheless, the FDA approval was thought to be impossible. At the time, the FDA had not approved anything for brain tumors for over 25 years. I was also shocked to find out that even after our delivery system was approved by the FDA, Medicare would not pay for it. So, we had to go public and petition and get political support from the US Congress to get Medicare to make it available for patients. After overcoming other innumerable roadblocks, we eventually received approval both for recurrent and newly diagnosed disease.

Even after these milestones were achieved, patient access was occasionally limited by clinical trials precluding its use. This is surprising because even recent studies performed in 2023 in France with 1600 patients at 42 institutions have shown statistically significant survival improvements and concluded that carmustine implanted biodegradable wafers should be a standard treatment of malignant gliomas.<sup>24</sup> The journey to achieve a real innovative change in clinical practice is resplendent with obstacles!

We have studied many different approaches, targets, and delivery devices over the last 40 years, and I am optimistic that many of them will ultimately, significantly affect and improve patient care.

## CONCLUSION

In conclusion, research progress is not the same as surgery. In surgery, when I take out a brain tumor, that patient goes home, it is wonderful, and the patient benefits immediately from our actions. Progress in the laboratory is incremental, but the progress only happens if you stay focused and persist with relentless effort.

The Clinician-Scientist legacy of Harvey Cushing and Walter Dandy (Figure 12) lives on at Hopkins and is vibrant throughout our extraordinary specialty of neurosurgery. We have an incredible group of people at Hopkins (Figure 13) continuing this tradition built by the giants, and it is not a ‘one man show’ like it was in Harvey Cushing’s and Walter Dandy’s time. This is an incredible group of devoted people all with the same mission: that of helping people in the maximum way through clinical work and through research. For me, the greatest satisfaction has been in mentoring the next generations to both master our exciting field and to advance it significantly for the patients.

From the day I started my own ongoing journey in neurosurgery, I am grateful for the privilege I have had in working with great people both as a student and then throughout my career with incredible colleagues, brilliant students, residents, and faculty. And of course, the tremendous trust from our patients who have allowed me to take care of them in the best way that we know how—which is good, but never good enough. There is so much more work to be performed. Ultimately, it is the patients who give us the stimulation and motivation to move the needle forward.

Thank you.

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